SOFTWARE ENGINEER INTERVIEW

Software Engineering & System Simulation

Production Software Development · GPU LLM Serving Research

Yu-Jui Liang

Education & Professional Experience

Education

National Taiwan University
M.S. Candidate, Data Science Master's Degree Program
Expected Graduation: June 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

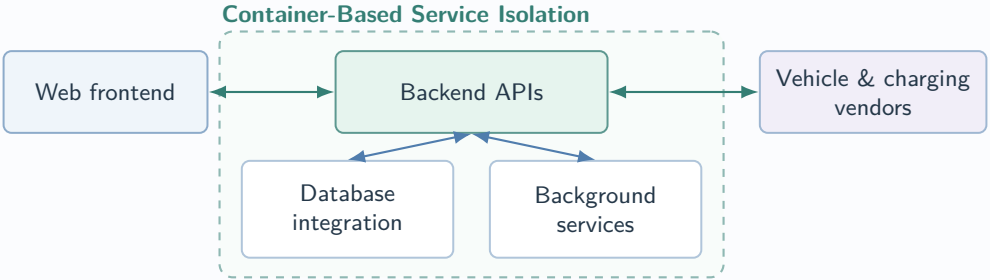
Professional Experience

AIMobile Co., Ltd.
Inventec Group
Contract Software Engineer
Apr. 2025 – Jul. 2025

Neopower Technologies Co., Ltd.
Contract Software Engineer
Jun. 2023 – Mar. 2025

NeoPower: Backend System Architecture

Electric Bus Charging Management Platform



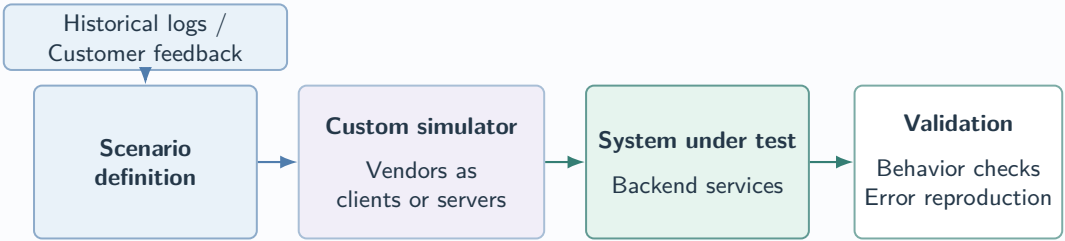
My contributions

System planning & backend development / Database integration

Log analysis & debugging / Deployment & maintenance

NeoPower: Scenario-Based System Simulation

I developed a custom simulator for vendor interaction and failure reproduction.



High Load

Many buses arrive at once

Network Failure

Mass disconnection of stations / chargers

NeoPower: Testing & Production Reliability

Pre-Deployment Testing

Simulator-based validation

Normal / abnormal conditions
Regression testing

Tests pass

Deploy updated system

Post-Deployment Validation

Charging-power allocation algorithms

Cross-Site Comparison: similar operating conditions

Some sites
Previous algorithm

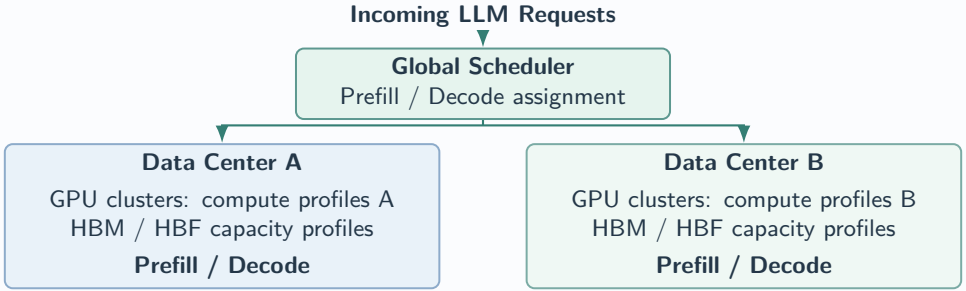
Other sites
Updated algorithm

Compare operating behavior

System logs / operating data
Customer feedback

Heterogeneous GPU LLM Serving Across Data Centers

Research objectives: higher serving throughput and lower total power

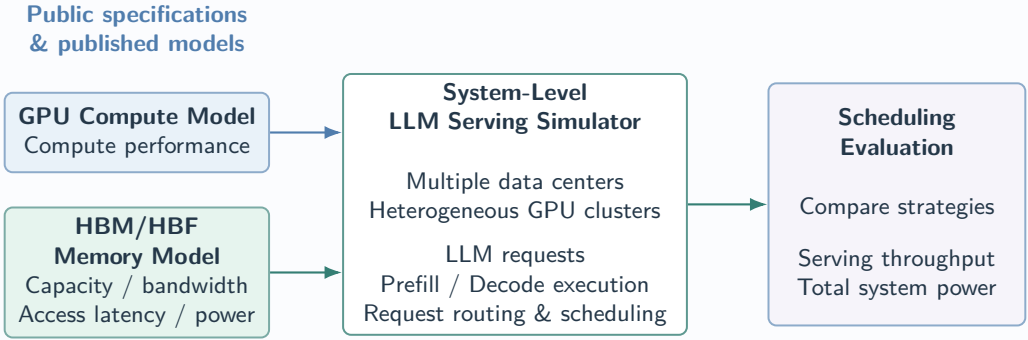


Memory configurations in the model

Full-HBM	**Full-HBF**	**Hybrid HBM/HBF**
Conventional baseline	Conceptual configuration	Conceptual example 50/50 capacity

Research Methodology: System-Level Simulation

Model heterogeneous hardware to evaluate LLM serving scheduling.



Architectural starting point: open-source GPU-cluster serving simulator

Planned integration: published GPU-memory and storage-system models

Thank You

Questions & Discussion